



Silicon Avalanche Photodiode KFDL-500

KFDL-series

Description

The KFDL-500 is a silicon-based avalanche photodiode with a peak sensitivity of 850 nm and a photosensitive area diameter of 500 μm .



Features

- High speed
- Multiplication gain >100
- 500 μm diameter active area
- Wide operating temperature range

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Applications

- Detection of low optical signals
- LIDAR and rangefindings
- Spectroscopy
- Optical communication system

Generic Characteristics at 25 °C

	Min	Typ	Max
Wavelength range, nm	400	-	1100
Peak sensitivity, nm	800	850	1000
Diameter of sense area, μm	-	500	-

Electrical Characteristics at 25 °C

	Min	Typ	Max
Breakdown voltage ($I_d=10\ \mu\text{A}$), V	120	160	220
Responsivity (905 nm), A/W	-	50	-
Gain ($I_d = 1\ \mu\text{A}$)	80	100	-
Breakdown voltage temperature coefficient	0.8	1.2	-
Dark current (Gain=10), nA	-	1	10
Capacitance, pF	-	2	3
Bandwidth (1 MHz, Gain=10), GHz	0.6	0.7	-

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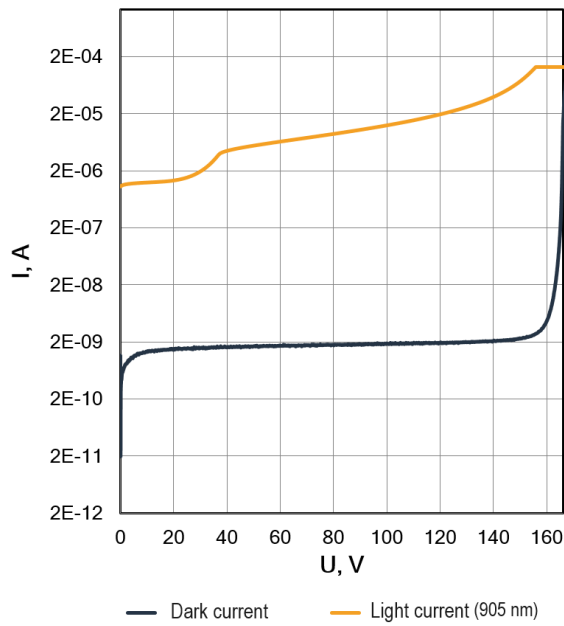


Fig 1. Typical Dark and Light Characteristic

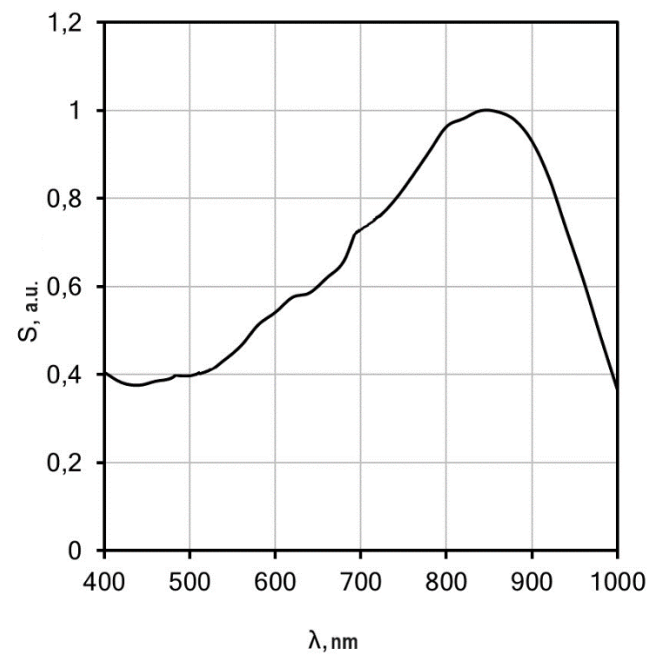


Fig 2. Spectral Response

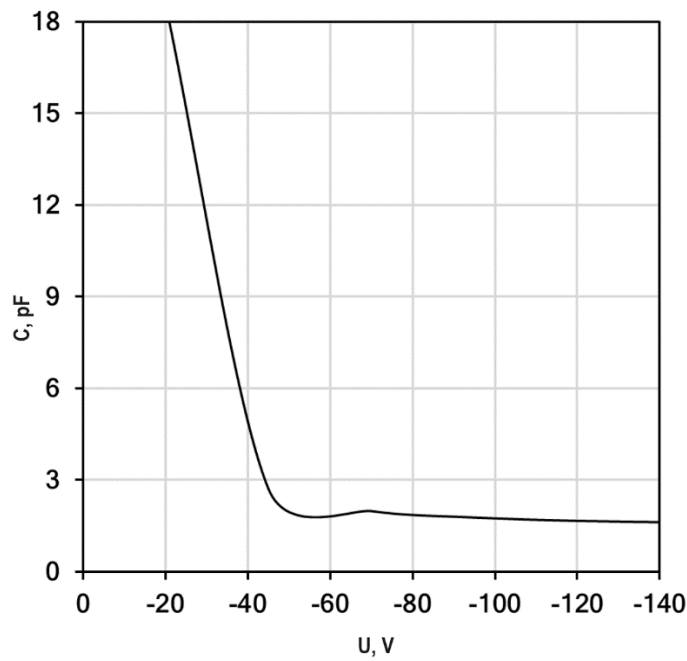


Fig 3. Capacitance vs Reverse Voltage (1 MHz)

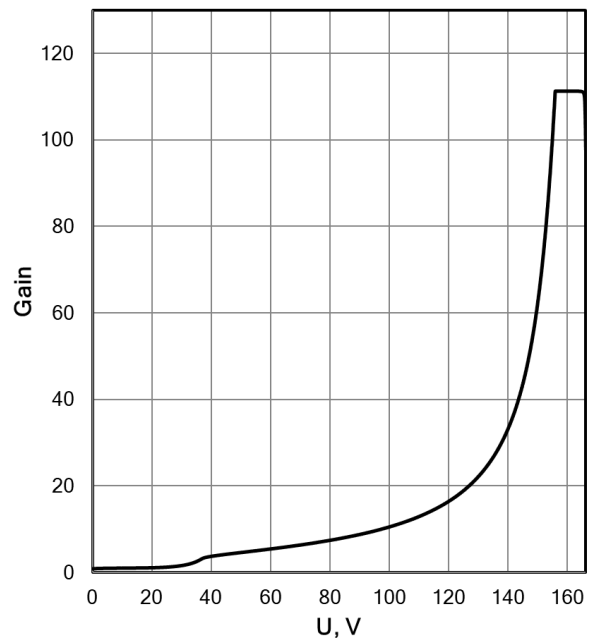


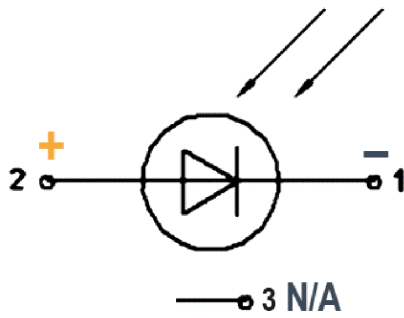
Fig 4. Gain vs Reverse Voltage (1 μ W, 905 nm)

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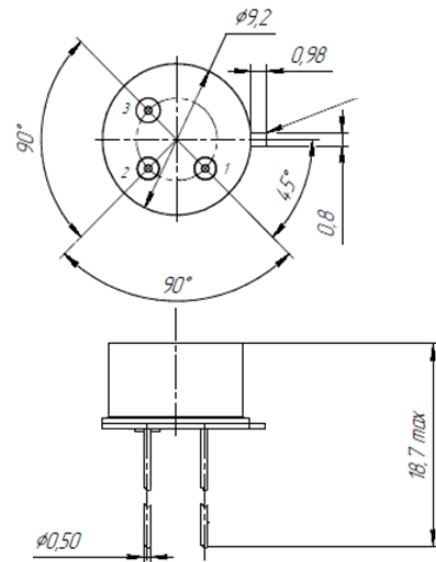


Package drawings

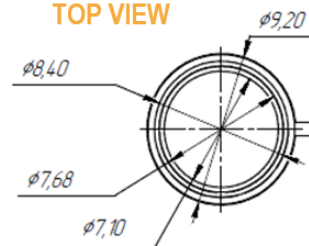
KFDL01A (TO-5 , 3-pin)



BOTTOM VIEW



TOP VIEW



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